

PSX Game Template – Documentation

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1. Project Overview

This is a **first-person PSX-style game template** built in Unreal Engine 5, designed for rapid prototyping of survival/adventure gameplay. Key systems included are:

- **Inventory system:** Stackable/non-stackable items with weight and slot management.
 - **Interactables:** Pickup items, storage containers, doors.
 - **PSX shader:** Retro visual effects including pixelation, color banding, VHS effects, fog, vertex wobble, and affine warping.
 - **Basic animations:** First-person movement, item usage, interactable interactions.
 - **Main menu:** Basic UI navigation.
 - **Audio setup:** Sound classes, sound mixes (e.g., dialogue and main), and basic sound effects for actions and item interactions.
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2. Systems

2.1 Inventory System

Purpose: Manage player items, stacking, quantities, and weight limits.

Key Components:

- **UInventoryComponent:** Handles item storage, adding/removing items, stack management, and total weight.
- **UItemBase:** Base item class.
- **UItemConsumable:** Consumable item type derived from **UItemBase**.

- **ID_Consumables** (Blueprint): Used to expand/add new items, setting mesh, item properties, and other metadata.

Features:

- Stackable and non-stackable items.
 - Splitting stacks and transferring between inventories (e.g., storage).
 - Integrates with player character for pickups, drops, and usage.
 - Item usage triggers sound and animation events.
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2.2 Interaction System

Purpose: Enable player to interact with objects in the world.

Key Components:

- **UInteractionComponent**: Handles line trace detection, focus, and timed interactions.
- **IInteractionInterface**: Interface for interactable actors.
- **APickupBase**: Pickups in the world.
- **AStorageBase**: Storage containers with inventory panels.
- **ADoorBase**: Doors with timeline-driven animations.

Features:

- Dynamic interaction prompts via HUD.
- Support for instant or timed interactions.
- Handles world focus for highlighting objects.
- Integrates with inventory for pickups and storage.

2.3 Player & Controller

Purpose: Manage first-person character movement, camera, and input.

Key Components:

- **APlayerCharacter**: First-person character with camera, inventory, and interaction component.
- **APlayerCharacterController**: Handles Enhanced Input actions such as movement, look, jump, crouch, sprint, interact, and toggling inventory.

Features:

- First-person movement with walking, sprinting, crouching, and jumping.
- Camera control with spring arm.
- Inventory and storage UI integration.
- Plays animations and sounds for item use and interactions.

2.4 HUD & UI

Purpose: Display player inventory, storage panels, and interaction prompts.

Key Components:

- **APlayerHUD**: Handles all UI elements.
- **PlayerWidget**: Main player HUD including inventory.
- **InteractionWidget**: Shows interaction prompts.
- **StoragePanel**: Displays container inventories.

Features:

- Show/hide inventory and storage panels.
 - Updates interaction prompts dynamically.
 - Integrates input mode switching for UI interaction.
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2.5 Audio Setup

Purpose: Basic audio system for gameplay sounds and UI feedback.

Setup:

- Sound classes for main game audio and dialogue.
- Sound class mixes to handle layered audio.
- Sound effects for pickups, item usage, and interactions.

Features:

- Configurable per-item and per-action sounds.
 - Can be extended to 3D positional audio.
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2.6 PSX Shader

Purpose: Provide a retro PSX-style visual aesthetic.

Features:

- Low-resolution pixelation.
- Color banding.
- VHS effects (scanlines, noise).
- Fog effects.

- Vertex wobble.
 - Affine texture warping.
 - Post-process compatible for flexible camera effects.
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3. Gameplay Features

- First-person pickup and use of items.
 - Inventory and storage management with weight/slot limitations.
 - Interact with doors and storage containers.
 - Playable animations for usage and interaction events.
 - Retro PSX-style visuals.
 - Basic audio feedback for player actions.
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4. Extensibility

- **Adding new items:** Expand `ID_Consumables` in Blueprints, set mesh and item properties.
 - **Adding new interactables:** Implement `IInteractionInterface`.
 - **Shader effects:** Can be extended or modified via the PSX shader's parameters.
 - **UI & HUD:** `PlayerWidget`, `StoragePanel`, and `InteractionWidget` are modular and can be customized.
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5. Roadmap / TODO

- Implement multiple save slots system.
- Add NPC interactions and dialogue system.
- Merge with vehicle template (driving physics, traffic, vehicles).
- Expand audio system for environmental sounds.
- Polish PSX shader with additional visual effects.